Species Identification in Noisy Soundscapes

A Deep Learning Approach using Sound Event Detection

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CSE 329 - Machine Learning Sessional · BUET

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Outline

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Inference Pipeline

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Summary

Problem Statement

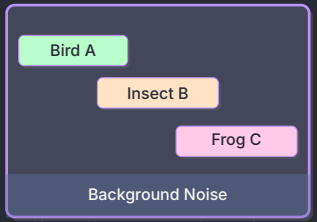
Acoustic Species Identification

Identify bird, insect, and amphibian species from **noisy soundscape recordings** in tropical rainforest environments.

Key Challenges:

- 206 different species to classify
- Heavy background noise (rain, wind)
- Overlapping species calls
- Extreme class imbalance
- 60-second long soundscape recordings

60s Soundscape



Dataset Overview

Training Data

- **Species**: 206 total (Aves, Amphibia)
- **Audio**: .ogg format (32 kHz)
- **Duration**: Variable (5s - 60s)

Class Imbalance

Problem: Species samples range from 1000+ to $<$ 10.

Test Data (Soundscapes)

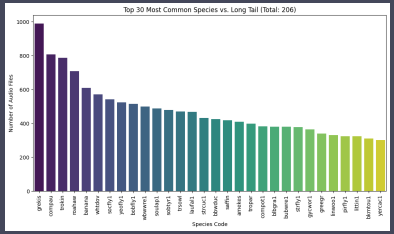
- 60-second continuous recordings
- Multiple species per recording
- Predictions per 5-second segment

Evaluation

Micro-averaged F1 Score · Top-1/5 Acc

Class Distribution: The Long-Tail Problem

Species Sample Counts



The Challenge

- **Head**: Common species have dense data.
- **Tail**: Rare species have sparse data (< 10).

Strategic Response

- **Weighted Random Sampling** ensures fair exposure to rare classes during training.

Audio Processing Pipeline

Raw Audio (20s, 32kHz) → Mel Spec (224 × 512) → Norm. (Z+MinMax) → 3-Channel Replication → CNN Input (3×224×512)

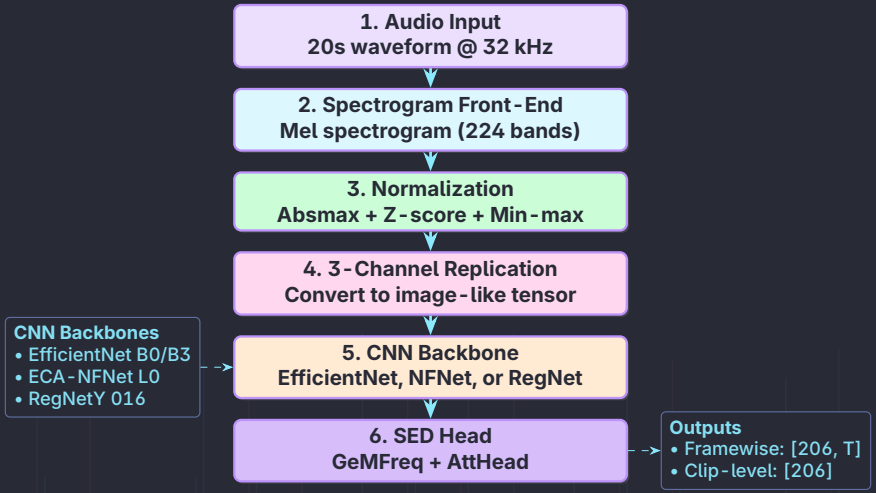
Mel Spectrogram

- **Mel Bands:** 224 · **n_fft:** 4,096
- **Hop Length:** 1,252
- **Freq Range:** 0 - 16 kHz

Why 20s Chunks?

- **Accuracy:** 87.2% (vs 84.2% @ 5s)
- Captures full temporal context
- Handles long Amphibia calls

Sound Event Detection (SED) Architecture



Key Architecture Components

GeMFreq Pooling

Generalized Mean Freq Pooling

- Formula: $\left(\frac{1}{|F|}\sum_f x_f^p\right)^{1/p}$
- **Learnable** parameter p
- Adapts: Max vs Average pooling
- Captures fine-grained freq patterns

Features Only

Preserves temporal dimension by removing classification head from backbone.

AttHead (Attention)

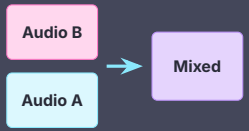
1. **Attention**: 1×1 Conv $\rightarrow$ Softmax
2. **Logits**: 1×1 Conv $\rightarrow$ 206
3. **Aggregation**:

$$\text{clip_logits} = \sum_t \text{att}_t \cdot \text{logits}_t$$

Benefit: Learns *when* species call!

Data Augmentation Techniques

Mixup



$$\tilde{x} = \lambda x_a + (1 - \lambda) x_b$$

+2.5% Acc

SpecAug

- Time: 20% masked
- Freq: 10% masked

+1.5% Acc

Sampling

Address imbalance:

$$w_c = \frac{1}{\sqrt{n_c}}$$

- Rare species ↑ oversampled
- Common species ↓

Essential

Training Configuration

Optimizer & Scheduler

- **Optimizer**: AdamW
  - LR: 5×10^{-4} · WD: 10^{-4}
- **Scheduler**: CosineAnnealing
  - $T_0 = 5$ epochs
  - $\eta_{min} = 10^{-6}$
- **Grad Clip**: max_norm = 1.0

Loss Function

CE + Label Smoothing

$$\mathcal{L} = -\sum_c y'_c \log(p_c)$$

- **Smoothing**: $\epsilon = 0.1$
- Prevents overconfidence

Setup

15-35 Epochs · Batch 64 · 5-Fold CV

Supervised Training Pipeline

1. Labeled Dataset
20s chunks with species labels

↓

2. Training Augmentation
MixUp and SpecAugment

↓

3. Weighted Sampling
Oversample rare species

↓

4. Forward Pass
Run SED → clip-level logits

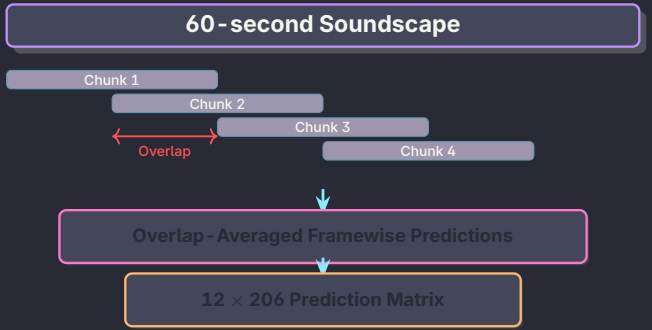
↓

5. Optimization
CE + AdamW + Grad Clipping

↓

6. Model Checkpoint
Save best validation base

Overlap-Average Inference



Process

12 overlapping chunks → average overlapping frame predictions.

Benefits

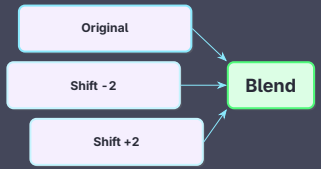
Robustness · TTA effect · **+0.3% Acc**

Refinement: TTA & Smoothing

Delta-Shift TTA

Temporal shift blending:

$$\text{pred} = 0.50 \cdot p_0 + 0.25 \cdot p_{\pm 2}$$



Gaussian Smoothing

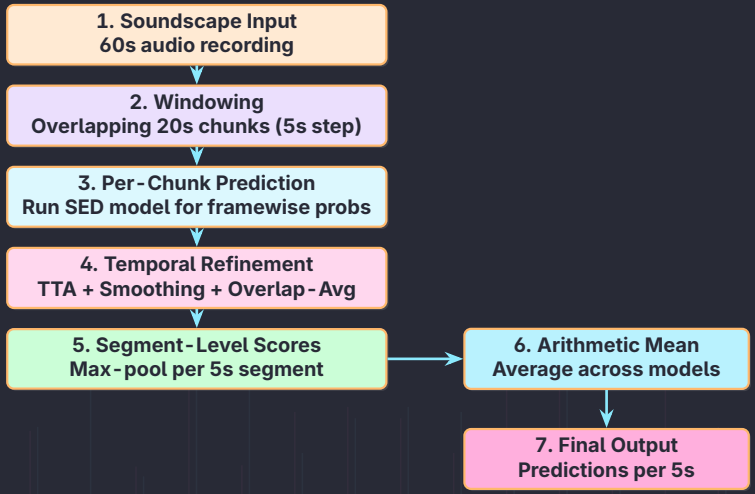
1D convolution to reduce jitter.

Kernel: $[0.1, 0.2, 0.4, 0.2, 0.1]$

Outcome

Free accuracy boost through temporal stability.

Inference & Ensemble Pipeline



Enabling Multi-Label Classification

From Single-Label to Multi-Label

Trained on primary labels, but predicts multiple concurrent species:

Architecture

- **Sigmoid Activation**: Independent binary tasks.
- **SED Localization**: Time-frequency bin evidence.

Inference Logic

- **Class Independence**: No forced competition.
- **Thresholding**: Score $> 0.3 \rightarrow$ Positive.

Detects 0, 1, or many species in a single audio chunk.

6 - Model Ensemble Architecture

Model	Backbone	Feat Dim	Stage
1	tf_efficientnet_b3	384	Trained
2	tf_efficientnet_b0	320	Trained
3	eca_nfnet_l0	2304	Trained
4	tf_efficientnet_b4	448	Pretrained
5	regnety_016	888	Pretrained
6	regnety_008	768	Pretrained

Strategy

Arithmetic Mean of sigmoid outputs.

Why?

Error Decorrelation · Robustness.

Performance Results

Model	Top-1 Acc	Top-5 Acc	Gain
Random Baseline	0.49%	2.43%	1×
Base (EfficientNet-B3)	60.07%	80.34%	122×
6-Model Ensemble	**87.99%**	**95.99%**	**180×**

Achievements

+28% from ensemble · **181×** vs random.

Contributions

Augmentation (+5%) · Ensemble (+3%) · TTA (+0.5%)

Sample Audio Predictions

Audio File	True	Top Pred	Conf.
XC254589	amekes	amekes	84.3%
XC493187	ywcpar	ywcpar	53.0%
XC449442	savhaw1	savhaw1	22.8%
XC171396	smbani	strher	11.7%

Observation

High confidence for clear calls; confusions in acoustic similarity.

All Techniques Implemented

Architecture:

1. SED (GeMFreq + AttHead)
2. 20s chunks, 224 mel bands
3. Z-score + min-max norm
4. Absmax audio norm

Training:

5. Mixup augmentation
6. SpecAugment
7. Weighted Random Sampling
8. Stochastic Depth (DropPath)
9. Multi-Iterative Noisy Student
10. CE + Label Smoothing
11. AdamW + CosineAnnealing

Inference:

12. Overlap-Average Inference
13. Delta-Shift TTA
14. Gaussian Smoothing
15. Multi-Model Ensemble

15 Techniques

Conclusion

Key Takeaways

- **87.99%** Accuracy achieved.
- 15 Advanced techniques integrated.
- Robust SED architecture.
- 6-Model Diverse Ensemble.

Impact

Biodiversity Monitoring · Automated Field Surveys

Thank You!

Questions?